43. In a railway reservation office, two clerks are engaged in checking reservation forms. On an average, the first clerk (A_1) checks 55% of the forms, while the second clerk (A_2) checks the remaining. A_1 has an error rate of 0.03 and A_2 has an error rate of 0.02. A reservation form is selected at random from the total number of forms checked during the day and is discovered to have some errors. Find the probability that the form is checked by A_1 and A_2 respectively.

44. In a university, 30% of the students are doing a course in statistics use the book authored by A_1, 45% use the book authored by A_2 and 25% use the book authored by A_3. The proportion of the students who learnt about each of these books through their teacher are $P(A_1)$ =0.50, $P(A_2)$=0.30, and $P(A_3)$=0.20. One of the student selected at random revealed that he learned the book he is using through their teachers. Find the probability that the book used it was authored by A_1, A_2, and A_3 respectively.

45. A bolt manufacturing company has four machines *A*, *B*, *C* and *D* producing 20%, 15%, 25% and 40% of the total output respectively. 5%, 4%, 3% and 2% of their output (in the same order) are defective bolts. A bolt is chosen at random from the factory and is found defective what is the probability of getting a defective bolt.

46. A city is partitioned into districts *A, B, C* having 20 percent, 40 percent and 40 percent of the registered voters respectively. The registered voters listed as Democrats are 50 percent in A, 25 percent in B and 75 percent in C. A registered voter is chosen randomly in the city. Find the probability that the voter is a listed democrat.

ANSWERS

I.1. (*a*) 2. (*b*) 3. (c) 4. (a) 5. (a) 6. (d) 7. (b) 8. (b) 9. (b) 10. (c)

II. 11. one 12. mutually exclusive 13. $\frac{1}{7}$ 14. $P(A_1) + P(A_2)$

15. less than or equal to 16. sure event 17.impossible event

18. A.N. Kolmogorov 19. $\frac{1}{3}$ 20. greater than

III. 21. 0.2 22. $\frac{2}{7}$ 23. $\frac{1}{2}$ 24. $\frac{3}{7}$ 25. $\frac{4}{9}$

IV.32. $\frac{2}{7}$ 33. $\frac{1}{6}$ 34. $\frac{5}{28}$ 35.(i) $\frac{1}{6}$ (ii) $\frac{5}{6}$ 36. 0.86 40. (i) $\frac{1}{3}$ (ii) $\frac{1}{4}$

V.42. $\frac{71}{120}$ 43. 0.647, 0.353 44. 0.45, 0.40, 0.15 45. $\frac{63}{2000}$ 46. 50%